

Cloud-Based IoT Health Monitoring System

A Secure, Offline-Resilient, Noise-Aware and Explainable Edge-Cloud Anomaly Detection Framework



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PROJECT GUIDE

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From Dashboard Concept to Trustworthy Monitoring

REVIEW-1 IMPROVEMENTS

ORIGINAL DIRECTION (PROPOSAL)

Generic Cloud-Based IoT Health Monitoring

Focus on sensor acquisition, cloud upload, and dashboard visualization. Assumed continuous connectivity and reliable sensor data. Primary goal was a functional data pipeline from Raspberry Pi to AWS/Azure.



REFINED DIRECTION (REVIEW-2)

Secure, Offline-Resilient, Noise-Aware Framework

Shift to a **local-first** architecture. The edge node (Raspberry Pi) is now responsible for evidence preservation and first-level interpretation. The system is designed to handle real-world constraints: noisy sensors and intermittent networks.

NOISE AWARENESS

OFFLINE RESILIENCE

EXPLAINABLE ML

SECURITY BY DESIGN

System Requirements and Hardware Boundary

1. SYSTEM REQUIREMENTS & HARDWARE

FUNCTIONAL REQUIREMENTS

FR-01 Acquire time-stamped physiological and environmental measurements.

FR-03 Detect and mark missing, implausible, or noisy sensor values.

FR-08 Generate explainable alerts with reason codes and severity.

FR-09 Commit observations locally before attempting cloud synchronization.

FR-12 Ensure idempotent cloud ingestion using unique event identifiers.

Scope Boundary

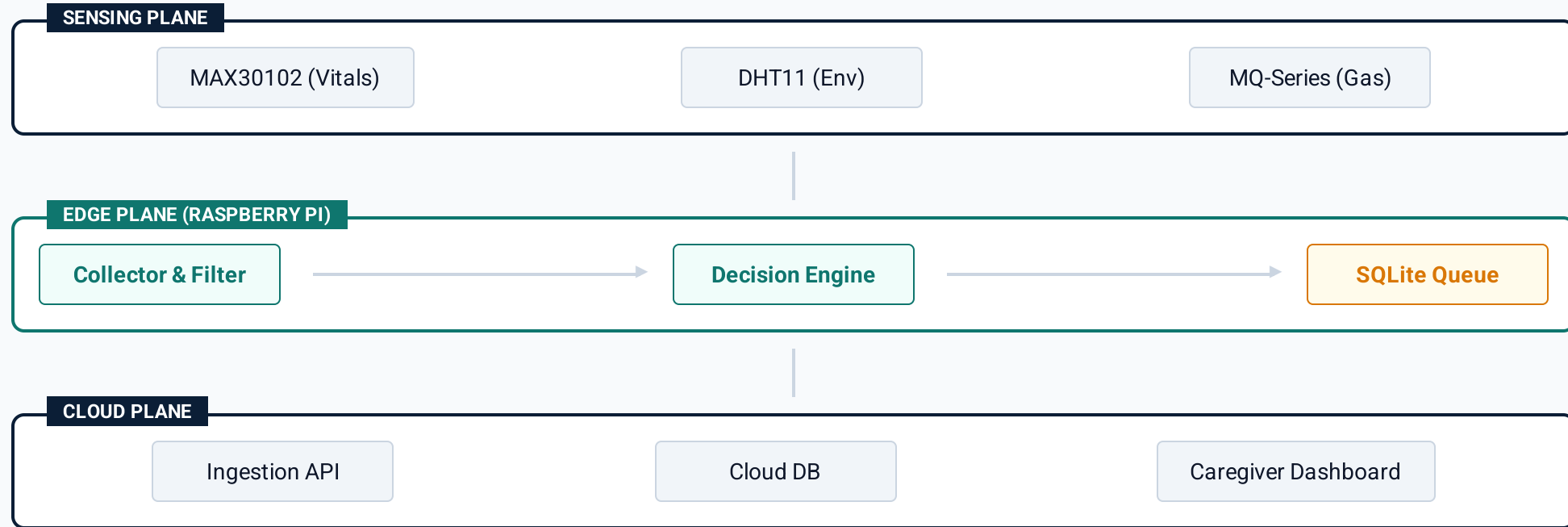
The prototype is a **monitoring and decision-support tool**. It is not a clinical diagnostic device. Thresholds are engineering parameters, not medical diagnoses.

HARDWARE STACK

Component	Purpose
Raspberry Pi 4	Edge Node / Gateway
MAX30102	Heart-Rate & SpO2
DHT11/MQ-Series	Environment
MicroSD 64GB	Local SQLite Store

Five-Plane Edge-Cloud Architecture

2. SYSTEM DESIGN & ARCHITECTURE



Security Plane: Device Identity • TLS Encryption • Authenticated API • Audit Logging

Noise-Aware and Explainable Decision Path

3. ALGORITHM & MODEL IMPLEMENTATION

1 Quality Gate

Checks missingness, physical plausibility, and short-window stability. Marks samples as INVALID, LOW, or ACCEPTABLE.

2 Robust Filtering

Applies a median filter over a short window to reduce isolated spikes while preserving raw data for audit.

3 Hybrid Decision

Combines transparent baseline rules with optional Isolation Forest scores. Generates reason codes and severity.

```
# Explainable Decision Record
{
  "quality_status": "ACCEPTABLE",
  "processed_value": 98.2,
  "baseline": 97.0,
  "deviation": 1.2,
  "severity": "NORMAL",
  "reason_codes": [],
  "explanation": "Within baseline."
}

# Anomaly Case
{
  "severity": "REVIEW",
  "reason_codes": ["deviation_from_baseline"],
  "explanation": "Decision based on: deviation_from_baseline."
}
```

The system prioritizes **transparency**: caregivers see exactly why an alert was raised.

Offline Resilience and Synchronization

System Operating States

NORMAL



DEGRADED (Offline)



RECOVERING



SAFE-STOP

Write-Before-Send Ordering

Observations are committed to the local SQLite queue **before** any network request. This ensures that a network failure cannot erase the event.

Idempotent Synchronization

Every record has a unique `event_id`. The cloud API honors this ID, preventing duplicate records during retries after reconnection.

Graceful Degradation

Loss of connectivity reduces remote visibility but does **not** stop local monitoring or erase evidence.

Edge Core Module

VERIFIED

```
class LocalStore:
    def add(self, event_id, sample, decision):
        self.db.execute("INSERT OR IGNORE...")
```

```
class HybridDetector:
    def evaluate(self, sample):
        status = self._quality(sample)
        processed = median(history)
```

```
$ python3 edge_monitor.py

[
  {
    "sensor": "spo2",
    "quality_status": "ACCEPTABLE",
    "sync_status": "PENDING"
  }
]

PY_COMPILE=PASS
```

Implementation Boundary

The current Review-2 artifact demonstrates the **hardware-agnostic edge core**. This verifies the data model, local persistence, and decision logic.

Not yet demonstrated:

- Physical sensor integration (MAX30102)
- Live cloud ingestion API
- Caregiver dashboard UI
- Isolation Forest model training

Team Roles and Review-1 Improvements

Member	Role	Review-2 Module
Palanati Kushal (2300031426)	HARDWARE INTEGRATION	Raspberry Pi & Sensor Interfacing
Y. Kishan (2300030979)	EDGE SOFTWARE	Local Persistence & Decision Engine
D. Varun (2300030156)	CLOUD PLATFORM	Ingestion API & Sync Logic
Dirishala Gopi Krishna (2300030916)	SYSTEM INTEGRATION	Architecture & Traceability

Key Improvements since Review-1

- Refined generic "cloud dashboard" into a failure-aware framework.
- Added explicit noise-awareness and quality-gating logic.
- Formalized offline resilience through write-before-send SQLite queue.
- Implemented explainable alerts with reason codes and severity.

Review-2 Test Matrix

INDIVIDUAL TEST RESULTS

ID	Scenario	Expected Result	Status
T-02	Event ID creation	Unique ID assigned before sync attempt	PASS
T-03	SQLite insertion	Observation is locally durable	PASS
T-04	Invalid sample	INVALID status; no high-confidence alert	PASS
T-06	Baseline deviation	Deviation and rule reason recorded	PASS
T-09	Write-before-send	Local record exists on network timeout	PASS
T-01	Sensor acquisition	Normalized sample from physical sensor	PENDING
T-10	Network outage	State becomes DEGRADED; queue grows	PENDING
T-12	Duplicate retry	Already-present event not duplicated	PENDING

Next Sprint: Physical sensor integration, Cloud API deployment, and Dashboard UI development.

Review-2 Ask: Demonstrate the Core, Defend the Boundary

Key Takeaways for the Panel

- ✓ We have refined the project into a failure-aware edge-cloud framework.
- ✓ The edge-core module is verified for quality-gating and local persistence.
- ✓ The system boundary is strictly non-diagnostic and engineering-led.
- ✓ We are ready to proceed with physical integration and cloud deployment.

Reference Repository: `github.com/krtaylor/IoT-Health-Monitor`